

Package ‘blstinypplot’

July 10, 2026

Title Plots for Seasonal Adjustment Analysts using tinypplot

Version 1.3

Description

Generates several types of time series plots useful for seasonal adjustment analysis using ggplot2. These routines rely heavily on the seasonal package to extract series and components from the 'seasonal adjustments generated by the US Census Bureau's X-13ARIMA-SEATS software, and can be generated from a single seas object or a list of seas objects. Types of plots include line plots, ratio plots, forecast plots, seasonal factor plots, seasonal adjustment component plots. Users can add grid lines, remove background shading, and shade recession regions in selected plots.

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tinypplot (>= 0.7),
tis

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absmax	<i>Maximum absolute value of a vector</i>
--------	---

Description

Generates the maximum of the absolute value of a numeric vector.

Usage

absmax(x = NULL, return_abs = TRUE)

Arguments

- x vector of numbers. This is a required element.
- return_abs Logical scalar; if TRUE, return the absolute maximum value. If FALSE, maintain the sign of the original value. Default is TRUE.

Details

Version 2.1, 2026-06-04

Value

Maximum of the absolute value of a vector.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

Examples

```
r50 <- rnorm(50)
r50.absmx <- absmx(r50)
```

add_outlier_lines	<i>Plot table from X-13ARIMA-SEATS seasonal adjustment run.</i>
-------------------	---

Description

Generate plot of user-specified table from the output of a seas object generated by the seasonal package.

Usage

```
add_outlier_lines(
  seas_obj = NULL,
  otl_col = NULL,
  add_sub_title = TRUE,
  sub_title_cex = 0.7
)
```

Arguments

seas_obj	A seas object for a single series generated from the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.
otl_col	Character array of length 6; color used for different outliers, with the order being 'ao', 'ls', 'tc', 'so', 'rp', 'tls'. Default is colors from the RColorBrewer palette "Paired" (Neuwirth 2022).
add_sub_title	Logical scalar; indicates if plots will include subtitles denoting what series are plotted. Default includes a subheader.
sub_title_cex	Numeric scalar; scaling for main title of plot. Default is 0.7.

Details

Version 1.4, 2026-07-08

Value

Outlier lines are added to an existing tinypplot plot.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Neuwirth E (2022). *RColorBrewer: ColorBrewer Palettes*. R package version 1.1-3, <https://CRAN.R-project.org/package=RColorBrewer>.

Sax C, Eddelbuettel D (2018). “Seasonal Adjustment by X-13ARIMA-SEATS in R.” *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
shoes_seas <-
  seasonal::seas(shoes2008, x11 = "",
    transform.function = "log",
    regression.aictest = c("td", "easter"),
    outlier.types = "all",
    arima.model = "(0 1 1)(0 1 1)",
    forecast.maxlead = 60)
shoes_sa <- seasonal::final(shoes_seas)
plot_series(shoes_sa,
  y_axis_label = "Shoes",
  do_grid = TRUE,
  draw_recess = TRUE,
  line_color = "black",
  start_plot = c(1995,1),
  this_point_type = 1,
  main_title = "X-11 Adjustment for Shoe Store Sales",
  main_title_line = 1.5,
  main_title_cex = 1.0,
  this_reset = TRUE)
add_outlier_lines(shoes_seas)
```

cnv_color_codes

Generates color names from hexadecimal input

Description

Generates vector of closest color names from hexadecimal color input. This uses the `color.id` function of the `plotrix` package (Lemon 2006) to do the conversion.

Usage

```
cnv_color_codes(color_vec = NULL)
```

Arguments

`color_vec` Vector of hexadecimal color codes. This is a required entry.

Details

Version 1.1, 2026-04-01

Value

vector of color names closest to hexadecimal color input.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Lemon J (2006). "Plotrix: a package in the red light district of R." *R-News*, **6**(4), 8-12.

Examples

```
Moonrise_Codes <-
  c("#F3DF6C", "#CEAB07", "#D5D5D3", "#24281A", "#798E87",
    "#C27D38", "#CCC591", "#29211F", "#85D4E3", "#F4B5BD",
    "#9C964A", "#CDC08C", "#FAD77B")
Moonrise_All <- cnv_color_codes(Moonrise_Codes)
```

color_blind_palette	<i>Color-blind friendly color palette</i>
---------------------	---

Description

Color palettes that can be used that can be distinguished by color-blind people (either from Neuwirth (2022) or Chang (2025)).

Usage

```
color_blind_palette(with_grey = TRUE)
```

Arguments

with_grey	Logical scalar; whether color blind palette contains 'grey', otherwise the palette contains black. Default is TRUE.
-----------	---

Details

Version 1.1, 2026-04-01

Value

Vector of hexadecimal color codes that form a color palette that can be distinguished by color-blind people.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Chang W (2025). "Colors (ggplot2) — cookbook-r.com." [http://www.cookbook-r.com/Graphs/Colors_\(ggplot2\)/](http://www.cookbook-r.com/Graphs/Colors_(ggplot2)/).

Neuwirth E (2022). *RColorBrewer: ColorBrewer Palettes*. R package version 1.1-3, <https://CRAN.R-project.org/package=RColorBrewer>.

Examples

```
line_color_blind <- color_blind_palette(FALSE)
```

convert_date_to_tis	<i>Convert ts dates to tis format</i>
---------------------	---------------------------------------

Description

Convert dates used for monthly (or quarterly) ts series to tis (Hallman 2021) formats.

Usage

```
convert_date_to_tis(
  this_date = NULL,
  this_freq = 12,
  is_start = TRUE,
  return_tis = FALSE
)
```

Arguments

this_date	Numeric scalar or vector; ts date to be converted. This is a required entry.
this_freq	Numeric scalar; frequency of ts time series. Default is 12 (monthly).
is_start	Logical scalar; date is assumed to be the beginning of the month. Default is TRUE; if FALSE date is assumed to be at the end of the month.
return_tis	Logical scalar; If TRUE, return as tis object; otherwise return as integer. Default is FALSE.

Details

Version 1.1, 2026-06-17

Value

A tis index value that is the equivalent of the ts date.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Hallman J (2021). *tis: Time Indexes and Time Indexed Series*. R package version 1.39, <https://CRAN.R-project.org/package=tis>.

Examples

```

this_month <- as.numeric(substr(Sys.Date(),1,4)) +
  (as.numeric(substr(Sys.Date(),6,7)) - 1) / 12
end_this_month_tis <-
  convert_date_to_tis(this_month, this_freq = 12,
    is_start = FALSE, return_tis = TRUE)
start_this_month_tis <-
  convert_date_to_tis(this_month, this_freq = 12,
    is_start = TRUE, return_tis = TRUE)

```

display_color	<i>Color name for display</i>
---------------	-------------------------------

Description

Generates color names for display on plot labels and subheaders.

Usage

```
display_color(line_color_code = NULL, strip_numbers = FALSE)
```

Arguments

line_color_code	Character string of color code to be used in plot. This is a required entry.
strip_numbers	Logical scalar that controls if numbers at the end of the text are stripped from the color name. Default is TRUE.

Details

Version 1.1, 2026-04-07

Value

Character string of color name closest to hexadecimal color input (if used); stripped of numbers if strip_numbers = TRUE.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

Examples

```

line_color_blind <- color_blind_palette()
line_color_blind_text <- array(NA, dim = 8)
for (i in 1:8) {
  line_color_blind_text[i] <- display_color(line_color_blind[i])
}

```

draw_recession	<i>Draw NBER recessions</i>
----------------	-----------------------------

Description

Draws shaded areas in plots corresponding to NBER recessions (NBER 2026).

Usage

```
draw_recession(
  line_color_recess = NULL,
  this_density = 50,
  this_border = NA,
  this_add_recess_start = NULL,
  this_sub_recess = TRUE,
  this_sub_line = 2.5,
  this_sub_cex = 0.75,
  display_color_strip = FALSE,
  convert_to_Date = FALSE,
  this_freq = 12
)
```

Arguments

<code>line_color_recess</code>	Character string; color used for shading recession periods. Default is 'lightgrey'. For a listing of R color codes, see R Coder (2025).
<code>this_density</code>	The density of shading lines, in lines per inch. The default value is 50. A zero value of density means no shading lines whereas negative values (and NA) suppress shading (and so allow color filling).
<code>this_border</code>	Integer scalar; thickness of border around region. Default is NA, meaning the border is not generated.
<code>this_add_recess_start</code>	Numeric scalar; Starting date for an additional recession period at the end of the series. Default is not to add recession dates.
<code>this_sub_recess</code>	Logical scalar; indicates if x-axis label for recession is produced for this plot. Default is x-axis label is produced
<code>this_sub_line</code>	Integer scalar; position of subtitle of plot. Default is 2.5.
<code>this_sub_cex</code>	Numeric scalar; scaling for subtitle of plot. Default is 0.75.
<code>display_color_strip</code>	Logical scalar; indicates if the display color will be stripped of trailing numbers. Default is FALSE.
<code>convert_to_Date</code>	Logical scalar; convert <code>start_recess</code> and <code>end_recess</code> to Date objects. Default is TRUE.
<code>this_freq</code>	Numeric scalar; frequency of ts time series. Default is 12 (monthly).

Details

Version 2.0, 2026-07-08

Value

Shades recession dates in plots generated by `tinypplot` (McDermott et al. 2025).

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

McDermott G, Arel-Bundock V, Zeileis A (2025). *tinypplot: Lightweight Extension of the Base R Graphics System*. doi:10.32614/CRAN.package.tinypplot, R package version 0.6.0.

NBER (2026). “Business Cycle Dating.” <https://www.nber.org/research/business-cycle-dating>.

R Coder (2025). “Colors in R — r-charts.com.” <https://r-charts.com/colors/>. [Accessed 17-02-2026].

Examples

```
air_seas <-
  seasonal::seas(AirPassengers,
    arima.model = '(0 1 1)(0 1 1)',
    x11='', x11.save = 'd12')
plot_table(air_seas, 'd12', 'AirPassengers',
  do_grid = TRUE,
  draw_recess = FALSE,
  use_ratio = FALSE,
  add_sub_title = TRUE,
  line_color = 'forestgreen')
draw_recession(line_color_recess = 'lightblue',
  this_border = 1,
  this_sub_line = 2.25,
  this_sub_cex = 0.75)
```

employment_data_mts	<i>US Unemployment Series, four main components in an mts object</i>
---------------------	--

Description

An `mts` object of the four main components of US Employment expressed as time series objects that end in December, 2022.

Usage

```
data(employment_data_mts)
```

Format

An mts object with 4 time series elements in four columns:

n2000013 Employed Males 16-19
n2000014 Employed Females 16-19
n2000025 Employed Males 20+
n2000026 Employed Females 20+

employment_list	<i>US Employment Series, four main components in a list object.</i>
-----------------	---

Description

A list object of the four main components of US Employment expressed as time series objects that end in December, 2022.

Usage

```
data(employment_list)
```

Format

A list object with 4 time series elements:

n2000013 Employed Males 16-19
n2000014 Employed Females 16-19
n2000025 Employed Males 20+
n2000026 Employed Females 20+

flag_peak	<i>Flag visual significant peaks in spectra</i>
-----------	---

Description

Determine positions of visual significant peaks in spectra using the criteria of Soukup and Findley (1999).

Usage

```
flag_peak(seas_obj = NULL, spec_type = NULL, spec_freq_code = NULL)
```

Arguments

seas_obj A seas object for a single series generated from the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.

spec_type Character string; type of spectrum. Possible values are 'ori', 'irr', 'rsd', 'sa'.

spec_freq_code Character string; type of frequency being tested. Possible values are 's' or 't'.

Details

Version 1.0, 2026-04-07

Value

If visually significant peaks found, a numeric vector of the position of the peak frequencies. If no peaks found, 0.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Sax C, Eddelbuettel D (2018). “Seasonal Adjustment by X-13ARIMA-SEATS in R.” *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Soukup RJ, Findley DF (1999). “On the Spectrum Diagnostics Used by X-12-ARIMA to Indicate the Presence of Trading Day Effects after Modeling or Adjustment.” *Proceeding of the American Statistical Association, Business and Economic Statistics Section*, 144–149. <https://www.census.gov/content/dam/Census/library/working-papers/1999/adrm/rr9903s.pdf>.

Examples

```
air_seas <-
  seasonal::seas(AirPassengers, arima.model = '(0 1 1)(0 1 1)', x11='')
this_flagged_peak_seas <- flag_peak(air_seas, 'ori', 's')
this_flagged_peak_td <- flag_peak(air_seas, 'ori', 't')
```

gen_tick_positions	<i>Generate tick mark positions</i>
--------------------	-------------------------------------

Description

Generate tick mark position for x or y axis for base R plots and plots generated by tinyplot (McDermott et al. 2025). Uses the lubridate package Grolemund and Wickham (2011)).

Usage

```
gen_tick_positions(
  this_series = NULL,
  tick_interval = 2,
  is_date = FALSE,
  time_interval = "years",
  this_base = 1
)
```

Arguments

<code>this_series</code>	Numeric or date vector, the values used for the x-axis. This is a required entry.
<code>tick_interval</code>	Integer scalar; the value separating the tick marks. Default is 2.
<code>is_date</code>	Logical scalar; indicates if <code>this_series</code> is a vector of Date objects. Default is FALSE, <code>this_series</code> is numeric.
<code>time_interval</code>	Character scalar; the time units used to construct the x-axis if <code>this_series</code> is a Date object vector. Default is "years".
<code>this_base</code>	Numeric scalar; number used to integer divide <code>this_series</code> for numeric vectors. Default is 1.

Details

Version 1.2.1, 2026-06-18

Value

Generates values for major tick marks for the X and Y axis of plots.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Grolemund G, Wickham H (2011). "Dates and Times Made Easy with lubridate." *Journal of Statistical Software*, **40**(3), 1–25. <https://www.jstatsoft.org/v40/i03/>.

McDermott G, Arel-Bundock V, Zeileis A (2025). *tinyplot: Lightweight Extension of the Base R Graphics System*. doi:10.32614/CRAN.package.tinyplot, R package version 0.6.0.

Examples

```
air_time <- time(AirPassengers)
air_Date <-
  as.Date(paste(as.numeric(floor(time(AirPassengers))),
               as.numeric(cycle(AirPassengers)),
               "1", sep = "-"))

air_x_axis_time <-
  gen_tick_positions(air_time, tick_interval = 3)
air_x_axis_Date <-
  gen_tick_positions(air_Date, tick_interval = 3, is_date = TRUE)
air_y_axis <-
  gen_tick_positions(AirPassengers, tick_interval = 200,
                    this_base = 100)
```

`get_auto_outlier_string`*Get automatic outlier names*

Description

Get the names of outliers identified in the seas object for a single series.

Usage

```
get_auto_outlier_string(seas_obj = NULL, return_string = TRUE)
```

Arguments

<code>seas_obj</code>	A seas object for a single series generated from the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.
<code>return_string</code>	Logical scalar; if TRUE, results returned as a character string (default). Else, results returned as a character array.

Details

Version 1.1, 2026-06-09

Value

Character string (or array if `return_string` is TRUE) containing a summary of the outliers identified in the regARIMA model. If no regressors or automatic outliers in the model, the routine will return a blank character (or blank vector).

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
air_seas <-  
  seasonal::seas(AirPassengers, arima.model = "(0 1 1)(0 1 1)", x11="")  
this_auto_outlier <- get_auto_outlier_string(air_seas)
```

get_month_index	<i>Generate index of month abbreviation</i>
-----------------	---

Description

Process string of month abbreviation to return a numeric index .

Usage

```
get_month_index(this_month_string, full_name = FALSE)
```

Arguments

this_month_string	Character string; 3 character abbreviation of month.
full_name	Logical scalar; indicates if this is the full name of the month

Details

Version 2.1, 2026-06-15

Value

Index of month - 1 for 'Jan', 2 for 'Feb', etc. If results do not match, this function will return 0.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

Examples

```
this0t1 <- 'A02015.Jan'
thisCode <- 'A0'
thisPerChar <- substr(this0t1,nchar(thisCode)+6,nchar(this0t1))
thisPerIndex <- get_month_index(thisPerChar)
```

get_recession_dates	<i>Get NBER recession dates</i>
---------------------	---------------------------------

Description

Add shading for US NBER recession dates (NBER 2026). Recession dates are taken from the nberDates function of the tis package (Hallman 2021).

Usage

```
get_recession_dates(
  start_recess = NULL,
  end_recess = NULL,
  convert_to_Date = FALSE,
  add_recess_start = NULL,
  this_freq = 12
)
```

Arguments

start_recess	Numeric scalar; Starting date for plot. This is a required entry.
end_recess	Numeric scalar; Ending date for plot. This is a required entry.
convert_to_Date	Logical scalar; convert start_recess and end_recess to Date objects.
add_recess_start	Numeric scalar; Starting date for an additional recession period at the end of the series. Default is not to add recession dates.
this_freq	Numeric scalar; frequency of ts time series. Default is 12 (monthly).

Details

Version 2.2, 2026-07-08

Value

Starting and ending dates for NBER recessions within a span of data.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Hallman J (2021). *tis: Time Indexes and Time Indexed Series*. R package version 1.39, <https://CRAN.R-project.org/package=tis>.

NBER (2026). “Business Cycle Dating.” <https://www.nber.org/research/business-cycle-dating>.

Examples

```
plot_limits <- c(1949, 1961)
thisRec <-
  get_recession_dates(start_recess = plot_limits[1],
                     end_recess = plot_limits[2])
```

get_reg_string	<i>Get names of regressors</i>
----------------	--------------------------------

Description

Generate string of names for the regressors used in the model fit for a given series.

Usage

```
get_reg_string(seas_obj = NULL, xreg_names = NULL, exclude_user = FALSE)
```

Arguments

seas_obj	A seas object for a single series generated from the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.
xreg_names	Character vector with names of user defined regressors used in model. Default is NULL, no user defined regressors. Number of names in this vector should match number of user-defined regressors; if not, a warning message will be produced.
exclude_user	Logical scalar; determines if user defined regressors are included in the string of regression names output by the function. Default is FALSE.

Details

Version 1.1, 2026-04-07

Value

Character string containing a summary of the regressors in the regARIMA model. If no regressors in the model, the routine will return a blank character.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
air_seas <-
  seasonal::seas(AirPassengers,
    transform.function = "log",
    forecast.maxlead=36,
    check.print = c( "pacf", "pacfplot" ),
    x11 = "", slidingspans = "")
air_reg <- get_reg_string(air_seas)
```

get_udg_index

Index for entry in UDG list

Description

Return index for entry in UDG list returned by the udg() function of the seasonal package.

Usage

```
get_udg_index(udg_list = NULL, this_key = NULL)
```


Arguments

udg_list	List object generated by udg function of the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.
this_key	Keyword found in udg files generated by X-13ARIMA-SEATS. This is a required entry.

Details

Version 3.1, 2026-03-05

Value

An integer denoting which element in the udg output matches the key provided by the user. If there is no match, the function returns the number 0.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Sax C, Eddelbuettel D (2018). “Seasonal Adjustment by X-13ARIMA-SEATS in R.” *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
ukgas_seas <-
  seasonal::seas(UKgas,
    series.period = 4,
    transform.function = "log",
    arima.model = "(0 1 1)(0 1 1)",
    forecast.maxlead = 20,
    check.print = c( "pacf", "pacfplot" ),
    x11 = "")
ukgas_udg <- seasonal::udg(ukgas_seas)
ukgas_udg_index_a <-
  get_udg_index(ukgas_udg, this_key='a')
```

plot_double_spectrum	<i>Generate double spectrum plot of the original and seasonally adjusted series.</i>
----------------------	--

Description

Generate plot of spectrum of original series and seasonally adjusted series on same axis. For more information on spectral diagnostics, see Sections 6.1 and 7.17 of the X-13ARIMA-SEATS Reference Manual (U. S. Census Bureau 2025).

Usage

```
plot_double_spectrum(
  seas_obj = NULL,
  xaxis_bls = TRUE,
  main_title = "AR Spectrum",
  series_name = NULL,
  do_sub = TRUE,
  sub_top = TRUE,
  line_color = NULL,
  this_palette = "Paired",
  this_plot_cex = 0.8,
  this_lab_cex = NULL,
  this_main_cex = NULL,
  this_sub_cex = NULL,
  this_axis_cex = NULL,
  y_axis_main_ticks = NULL,
  main_title_line = 3,
  sub_title_line = NULL,
  display_color_strip = FALSE,
  do_frame = TRUE,
  this_reset = TRUE
)
```

Arguments

<code>seas_obj</code>	A seas object for a single series generated from the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.
<code>xaxis_bls</code>	Logical scalar; indicates if x-axis of spectral plot will be frequency by month rather than the actual frequencies. Default sets x-axis to frequency by month.
<code>main_title</code>	Character string; main title of plot. Default is 'AR Spectrum'.
<code>series_name</code>	Character scalar; name of the time series.
<code>do_sub</code>	Logical scalar; indicates if subtitle is generated. Default is to generate the subtitle.
<code>sub_top</code>	Logical scalar; if TRUE, subtitle will appear below the title. If FALSE, subtitle will appear below the x-axis. Default is TRUE.
<code>line_color</code>	Character array of length 6; color used in spectrum plots, in the order of spectrum of ori, spectrum of SA, line for seasonal frequency, line for TD frequency, star for visually significant seasonal frequency, star for visually significant TD frequency. Default is NULL, which indicates that the palette specified in <code>this_palette</code> is used to generate colors for this plot. For a listing of R color codes, see R Coder (2025).
<code>this_palette</code>	Colors used for lines in this plot. Default is the RColorBrewer palette "Paired" (Neuwirth 2022).
<code>this_plot_cex</code>	Numeric scalar; scaling for the plot itself. Default is 0.8.
<code>this_lab_cex</code>	Numeric scalar; scaling for plot labels. Default is the value of <code>this_plot_cex</code> .
<code>this_main_cex</code>	Numeric scalar; scaling for main plot title. Default is the value of <code>this_plot_cex</code> .
<code>this_sub_cex</code>	Numeric scalar; scaling for plot subtitle. Default is the value of <code>this_plot_cex</code> .
<code>this_axis_cex</code>	Numeric scalar; scaling for plot axis. Default is the value of <code>this_plot_cex</code> .

y_axis_main_ticks	Numeric vector; Y-axis tick marks. Default is determined by the axis function.
main_title_line	Integer scalar; position of main title of plot. Default is 3.00.
sub_title_line	Integer scalar; position of main title of plot. Default depends on the entry for sub_top; 1 if TRUE, 2 if FALSE.
display_color_strip	Logical scalar; indicates if the display color will be stripped of trailing numbers. Default is FALSE.
do_frame	Logical scalar; indicates the plot will have a frame. Default is the plot frame will be produced.
this_reset	Logical scalar; if TRUE, the values of par are reset. Default is TRUE.

Details

Version 1.5, 2026-07-08

Value

Generate plot of spectrum of original series and seasonally adjusted series on same axis.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Neuwirth E (2022). *RColorBrewer: ColorBrewer Palettes*. R package version 1.1-3, <https://CRAN.R-project.org/package=RColorBrewer>.

R Coder (2025). “Colors in R — r-charts.com.” <https://r-charts.com/colors/>. [Accessed 17-02-2026].

Sax C, Eddelbuettel D (2018). “Seasonal Adjustment by X-13ARIMA-SEATS in R.” *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

U. S. Census Bureau (2025). “X-13ARIMA-SEATS Reference Manual, Version 1.1.” U.S. Census Bureau, U.S. Department of Commerce (<https://www2.census.gov/software/x-13arima-seats/x13as/windows/documentation/docx13ashtml.pdf>).

Examples

```
air_seas <-
  seasonal::seas(AirPassengers,
    arima.model = '(0 1 1)(0 1 1)',
    x11='',
    spectrum.save = c("sp1", "sp0"))
plot_double_spectrum(air_seas,
  series_name = 'AirPassengers',
  line_color = c('blue', 'green', 'darkblue',
    'darkgreen', 'red', 'orange'))
```

plot_fcst

*Forecast plot***Description**

Generates a plot of the regARIMA forecasts with confidence bounds.

Usage

```
plot_fcst(
  seas_obj = NULL,
  main_title = "ARIMA forecasts",
  do_sub = TRUE,
  do_grid = FALSE,
  do_frame = TRUE,
  length_ori = 2,
  line_color = c("darkgrey", "blue", "darkgreen"),
  this_plot_cex = 0.8,
  this_lab_cex = NULL,
  this_main_cex = 1,
  this_sub_cex = 0.7,
  this_axis_cex = NULL,
  x_axis_label = "Time",
  y_axis_label = NULL,
  x_axis_main_ticks = NULL,
  y_axis_main_ticks = NULL,
  main_title_line = 2.25,
  sub_title_line = 0.75,
  this_reset = TRUE
)
```

Arguments

seas_obj	A seas object for a single series generated from the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.
main_title	Character string; main title of plot. Default is 'ARIMA Residuals'.
do_sub	Logical scalar; indicates if subtitle is generated. Default is to generate the subtitle.
do_grid	Logical scalar; indicates if certain plots will have grid lines. Default is no grid lines.
do_frame	Logical scalar; indicates the plot will have a frame. Default is the plot frame will be produced.
length_ori	Integer scalar; number of years of the original series to show with forecasts. Default is 2 years.
line_color	Array of character strings; color used for original series, forecast, and forecast bounds. Default is c("darkgrey", "blue", "darkgreen"). For a listing of R color codes, see R Coder (2025).
this_plot_cex	Numeric scalar; scaling for the plot itself. Default is 0.8.
this_lab_cex	Numeric scalar; scaling for plot labels. Default is the value of this_plot_cex.

<code>this_main_cex</code>	Numeric scalar; scaling for main plot title. Default is the value of 1.0.
<code>this_sub_cex</code>	Numeric scalar; scaling for plot subtitle. Default is the value of 0.7.
<code>this_axis_cex</code>	Numeric scalar; scaling for plot axis. Default is the value of <code>this_plot_cex</code> .
<code>x_axis_label</code>	Character string; x-axis label for plot, if specified.
<code>y_axis_label</code>	Character string; y-axis label for plot, if specified.
<code>x_axis_main_ticks</code>	Numeric vector; X-axis tick marks. Default is determined by the axis function.
<code>y_axis_main_ticks</code>	Numeric vector; Y-axis tick marks. Default is determined by the axis function.
<code>main_title_line</code>	Integer scalar; position of main title of plot. Default is 2.25.
<code>sub_title_line</code>	Integer scalar; position of main title of plot. Default is 0.75.
<code>this_reset</code>	Logical scalar; if TRUE, the values of <code>par</code> are reset. Default is TRUE.

Details

Version 1.6, 2026-07-10

Value

Generates a plot of the regARIMA forecasts with confidence bounds.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

R Coder (2025). “Colors in R — r-charts.com.” <https://r-charts.com/colors/>. [Accessed 17-02-2026].

Sax C, Eddelbuettel D (2018). “Seasonal Adjustment by X-13ARIMA-SEATS in R.” *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
air_seas <-
  seasonal::seas(AirPassengers,
    arima.model = '(0 1 1)(0 1 1)',
    forecast.maxlead = 60,
    forecast.save = "fct",
    series.save = "a1")
plot_fcst(air_seas,
  main_title = 'Forecasts for Airline Passengers',
  do_grid = TRUE)
```

plot_fcst_history	<i>Generate forecast history plot</i>
-------------------	---------------------------------------

Description

Generate forecast history plot, which compares the sum of squared forecast errors for two models.

Usage

```
plot_fcst_history(
  seas_md11 = NULL,
  seas_md12 = NULL,
  start_hist = NULL,
  main_title = "Differences in the Sum of Squared Forecast Errors",
  name_md11 = "Model 1",
  name_md12 = "Model 2",
  do_grid = FALSE,
  do_frame = TRUE,
  line_color = NULL,
  this_plot_cex = 0.8,
  this_lab_cex = NULL,
  this_main_cex = NULL,
  this_sub_cex = NULL,
  this_axis_cex = NULL,
  this_mar = c(5.1, 2.1, 5.1, 0.5),
  main_title_line = 2.5,
  sub_title_line = 1,
  display_color_strip = TRUE,
  x_axis_label = NULL,
  y_axis_label = NULL,
  x_axis_main_ticks = NULL,
  y_axis_main_ticks = NULL,
  this_reset = TRUE
)
```

Arguments

seas_md11	A seas object for a single series generated from the seasonal package (Sax and Eddelbuettel 2018) for the first model. This is a required entry.
seas_md12	A seas object for a single series generated from the seasonal package for the second model fit to the same series as for seas_md11. This is a required entry.
start_hist	integer scalar; starting date for the history analysis. Could be an array of length 2; will be converted to a scalar. This is a required entry.
main_title	Character string; main title of plot. Default is 'Differences in the Sum of Squared Forecast Errors'.
name_md11	Character string; Description of first model for use in the subtitle. Default is 'Model 1'.
name_md12	Character string; Description of second model for use in the subtitle. Default is 'Model 2'.

do_grid	Logical scalar; indicates if certain plots will have grid lines. Default is no grid lines.
do_frame	Logical scalar; indicates the plot will have a frame. Default is the plot frame will be produced.
line_color	Character array of length 2; color used for each forecast lag. Default is colors from the <code>rcartocolor</code> palette "Bold" (Nowosad 2018); this palette is extracted using the <code>paletteer</code> package (Hvitfeldt 2021). For a listing of R color codes, see R Coder (2025).
this_plot_cex	Numeric scalar; scaling for the plot itself. Default is 0.8.
this_lab_cex	Numeric scalar; scaling for plot labels. Default is the value of <code>this_plot_cex</code> .
this_main_cex	Numeric scalar; scaling for main plot title. Default is the value of <code>this_plot_cex</code> + 0.1.
this_sub_cex	Numeric scalar; scaling for plot subtitle. Default is the value of <code>this_plot_cex</code> .
this_axis_cex	Numeric scalar; scaling for plot axis. Default is the value of <code>this_plot_cex</code> .
this_mar	Numeric vector; set margins for the plot. Default is <code>c(5.1, 2.1, 5.1, 0.5)</code> .
main_title_line	Integer scalar; position of main title of plot. Default is 2.25.
sub_title_line	Integer scalar; position of main title of plot. Default is 1.
display_color_strip	Logical scalar; indicates if the display color will be stripped of trailing numbers. Default is TRUE.
x_axis_label	Character string; x-axis label for plot, if specified.
y_axis_label	Character string; y-axis label for plot, if specified.
x_axis_main_ticks	Numeric vector; X-axis tick marks. Default is determined by the <code>axis</code> function.
y_axis_main_ticks	Numeric vector; Y-axis tick marks. Default is determined by the <code>axis</code> function.
this_reset	Logical scalar; if TRUE, the values of <code>par</code> are reset. Default is TRUE.

Details

Version 1.4, 2026-07-10

Value

Generate forecast history plot. Can be more than one series. If series not specified, print out error message and return NULL.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Hvitfeldt E (2021). *paletteer: Comprehensive Collection of Color Palettes*. R package version 1.3.0, <https://github.com/EmilHvitfeldt/paletteer>.

Nowosad J (2018). *'CARTOColors' Palettes*. R package version 1.0.0, <https://jakubnowosad.com/rcartocolor/>.

R Coder (2025). “Colors in R — r-charts.com.” <https://r-charts.com/colors/>. [Accessed 17-02-2026].

Sax C, Eddelbuettel D (2018). “Seasonal Adjustment by X-13ARIMA-SEATS in R.” *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
air_seas_md1 <-
  seasonal::seas(AirPassengers, x11='',
    slidingspans = '', transform.function = 'log',
    arima.model = '(0 1 1)(0 1 1)',
    regression.aictest = NULL, outlier = NULL,
    forecast.maxlead = 36,
    check.print = c( 'pacf', 'pacfplot' ),
    history.fstep = c(1, 12), history.estimates = 'fcst',
    history.save = 'fcsterrors')
air_seas_md12 <-
  seasonal::seas(AirPassengers, x11='',
    slidingspans = '', transform.function = 'log',
    arima.model = '(0 1 1)(0 1 1)',
    regression.variables = c("td"),
    forecast.maxlead = 36,
    check.print = c( 'pacf', 'pacfplot' ),
    history.fstep = c(1, 12), history.estimates = 'fcst',
    history.save = 'fcsterrors')
plot_fcst_history(air_seas_md1, air_seas_md12,
  start_hist = 1957.0,
  main_title = 'Differences in the SS Fcst Error for Airline Passengers',
  name_md1 = 'Airline model',
  name_md12 = 'Airline model + regressors')
```

plot_fts

Final t-statistics for the outlier identification procedure plot

Description

Generates a plot of the final t-statistics for the outlier identification procedure.

Usage

```
plot_fts(
  seas_obj = NULL,
  fts = NULL,
  this_cex = 0.5,
  start_plot = NULL,
  main_title = "Outlier T-Values",
  add_identified_outliers = FALSE,
  do_grid = FALSE,
  outlier_color = NULL,
  this_palette = "Dark2",
  this_plot_cex = 0.8,
```



```

    this_lab_cex = NULL,
    this_main_cex = NULL,
    this_sub_cex = NULL,
    this_axis_cex = NULL,
    this_mar = c(4, 4, 4, 0.5),
    main_title_line = 2.25,
    sub_title_line = 1,
    x_axis_label = NULL,
    y_axis_label = NULL,
    x_axis_main_ticks = NULL,
    y_axis_main_ticks = NULL,
    plot_abs = TRUE,
    this_reset = TRUE
  )

```

Arguments

seas_obj	A seas object for a single series generated from the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.
fts	time series matrix containing final outlier t-statistics for all types of outlier specified by the user. This is a required entry.
this_cex	Numeric scalar; sets the cex plotting parameter. Default sets $cex = 0.5$.
start_plot	Integer vector of length 2; Starting date for plot. Default is starting date for the time series.
main_title	Character string; main title of plot. Default is 'Outlier T-Values'.
add_identified_outliers	Logical scalar; indicates if outlier plots will include identified outliers. Default is not including identified outliers.
do_grid	Logical scalar; indicates if certain plots will have grid lines. Default is no grid lines.
outlier_color	Character array of length 3; color used for different outliers, with the order being 'ao', 'ls', 'tc'. Default is NULL, which indicates that the palette specified in this_palette is used to generate colors for this plot. For a listing of R color codes, see R Coder (2025).
this_palette	Character string; default RColorBrewer palette (Neuwirth 2022). Deault is "Dark2".
this_plot_cex	Numeric scalar; scaling for the plot itself. Default is 0.8.
this_lab_cex	Numeric scalar; scaling for plot labels. Default is the value of this_plot_cex.
this_main_cex	Numeric scalar; scaling for main plot title. Default is the value of this_plot_cex + 0.1.
this_sub_cex	Numeric scalar; scaling for plot subtitle. Default is the value of this_plot_cex.
this_axis_cex	Numeric scalar; scaling for plot axis. Default is the value of this_plot_cex.
this_mar	Numeric vector; set margins for the plot. Default is $c(4, 4, 4, 0.5)$.
main_title_line	Integer scalar; position of main title of plot. Default is 2.25.
sub_title_line	Integer scalar; position of main title of plot. Default is 1.
x_axis_label	Character string; x-axis label for plot, if specified.

y_axis_label	Character string; y-axis label for plot, if specified.
x_axis_main_ticks	Numeric vector; X-axis tick marks. Default is determined by the axis function.
y_axis_main_ticks	Numeric vector; Y-axis tick marks. Default is determined by the axis function.
plot_abs	Logical scalar; if TRUE, plot the maximum absolute value of the t-stats. If FALSE, maintain the sign of the original t-statistic in the plot. Default is TRUE.
this_reset	Logical scalar; if TRUE, the values of par are reset. Default is TRUE.

Details

Version 2.5, 2026-07-10

Value

Generates a plot of the final t-statistics from the automatic outlier identification procedure.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Neuwirth E (2022). *RColorBrewer: ColorBrewer Palettes*. R package version 1.1-3, <https://CRAN.R-project.org/package=RColorBrewer>.

R Coder (2025). “Colors in R — r-charts.com.” <https://r-charts.com/colors/>. [Accessed 17-02-2026].

Sax C, Eddelbuettel D (2018). “Seasonal Adjustment by X-13ARIMA-SEATS in R.” *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
air_seas_outlier <-
  seasonal::seas(AirPassengers,
    arima.model = '(0 1 1)(0 1 1)',
    outlier.types = 'all',
    outlier.save = "fts")
air_fts_good <- seasonal::series(air_seas_outlier, "fts")
plot_fts(air_seas_outlier, air_fts_good,
  main_title = 'Outlier T-Values for Airline Passengers')
plot_fts(air_seas_outlier, air_fts_good,
  main_title = 'Outlier T-Values for Airline Passengers',
  add_identified_outliers = TRUE)
plot_fts(air_seas_outlier, air_fts_good, plot_abs = FALSE,
  main_title = 'Outlier T-Values for Airline Passengers',
  add_identified_outliers = TRUE)
```

plot_ratio	<i>Ratio plot</i>
------------	-------------------

Description

Generates a high-definition plot around a reference line other than zero.

Usage

```
plot_ratio(
  ratio_series = NULL,
  ratio_range = range(ratio_series),
  main_title = "Ratio Plot",
  main_title_line = 2,
  main_title_cex = NULL,
  ratio_mean = 1,
  ratio_color = "black",
  ratio_mean_color = "black",
  x_axis_main_ticks = NULL,
  y_axis_main_ticks = NULL,
  x_axis_label = "Time",
  y_axis_label = NULL,
  this_plot_cex = 0.8,
  this_lab_cex = NULL,
  this_axis_cex = NULL,
  this_mar = c(4, 4, 4, 0.5),
  this_reset = TRUE
)
```

Arguments

ratio_series	Time series of ratios/factors for which you want to generate a high definition plot. This is a required entry.
ratio_range	Range of values you wish the plot to be plotted over. Default is range of the series.
main_title	Title for the plot. Default is character string 'Ratio Plot'.
main_title_line	Integer scalar; position of main title of plot. Default is 2.
main_title_cex	Numeric scalar; scaling for main title of plot. Default is this_plot_cex + 0.25.
ratio_mean	Assumed mean value for the ratio. Default is 1.0
ratio_color	Color used for lines in ratio plot. Default is 'black'.
ratio_mean_color	Color used for mean line in ratio plot. Default is 'black'.
x_axis_main_ticks	Numeric vector; X-axis tick marks. Default is determined by the axis function.
y_axis_main_ticks	Numeric vector; Y-axis tick marks. Default is determined by the axis function.
x_axis_label	Character string; x-axis label for plot. Default is 'Time'.

y_axis_label	Character string; y-axis label for plot, if specified.
this_plot_cex	Numeric scalar; scaling for the plot itself. Default is 0.8.
this_lab_cex	Numeric scalar; scaling for plot labels. Default is the value of this_plot_cex.
this_axis_cex	Numeric scalar; scaling for plot axis. Default is the value of this_plot_cex.
this_mar	Numeric vector; set margins for the plot. Default is c(4,4,4,0.5).
this_reset	Logical scalar; if TRUE, the values of par are reset. Default is TRUE.

Details

Version 1.5.2, 2026-06-03

Value

Generates a high definition plot of ratios centered on one, by default.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

Examples

```
air_seas <-
  seasonal::seas(AirPassengers,
    transform.function= 'log',
    arima.model = '(0 1 1)(0 1 1)',
    seats.save = 's10')
air_sf <- seasonal::series(air_seas, 's10')
plot_ratio(air_sf,
  main_title = 'SEATS seasonal for Airline Passenger',
  ratio_color = 'darkblue')
```

plot_resid	<i>Residual plot</i>
------------	----------------------

Description

Generates a plot of the regARIMA residuals with diagnostic information.

Usage

```
plot_resid(
  seas_obj = NULL,
  main_title = "ARIMA Residuals",
  main_title_line = 2.75,
  main_title_cex = 1,
  series_name = NULL,
  x_axis_label = NULL,
  y_axis_label = NULL,
  x_axis_main_ticks = NULL,
  y_axis_main_ticks = NULL,
  do_grid = TRUE,
```

```

draw_recess = FALSE,
recess_start = NULL,
recess_col = NULL,
recess_sub = TRUE,
this_trans = TRUE,
use_ratio = FALSE,
line_color = "green",
this_plot_cex = 0.8,
this_lab_cex = NULL,
this_axis_cex = NULL,
this_sub_cex = 0.5,
this_mar = c(4, 4, 4, 0.5),
sub_title_one_line = 1.5,
sub_title_two_line = 0.5,
this_reset = TRUE
)

```

Arguments

seas_obj	A seas object for a single series generated from the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.
main_title	Character string; main title of plot. Default is 'ARIMA Residuals'.
main_title_line	Integer scalar; position of main title of plot. Default is 2.75.
main_title_cex	Numeric scalar; scaling for main title of plot. Default is 1.25.
series_name	Character scalar; name of the time series used in seas_obj. Used only if specified.
x_axis_label	Character string; x-axis label for plot, if specified.
y_axis_label	Character string; y-axis label for plot, if specified.
x_axis_main_ticks	Numeric vector; X-axis tick marks. Default is determined by the axis function.
y_axis_main_ticks	Numeric vector; Y-axis tick marks. Default is determined by the axis function.
do_grid	Logical scalar; indicates if certain plots will have grid lines. Default is grid lines plotted.
draw_recess	Logical scalar; indicates if certain plots will have shaded areas for NBER recession dates (NBER 2026). Default is recession shading not plotted.
recess_start	numeric matrix; Rows of dates for additional recession starting and ending dates. Default is not to add recession dates.
recess_col	Character string; color used for shading of recession region. Default is 'lightgrey'.
recess_sub	Logical scalar; indicates if x-axis label for recession is produced for this plot. Default is x-axis label is produced
this_trans	Logical scalar; indicates if the adjustment was done with a log transform. Default is TRUE.
use_ratio	Logical scalar; indicates if plots of seasonal factors, irregular, and residuals are done as ratio plots. Default has these plots as time series line plots.
line_color	Character string; color used for residuals. Default is 'green'. For a listing of R color codes, see R Coder (2025).

this_plot_cex	Numeric scalar; scaling for the plot itself. Default is 0.8.
this_lab_cex	Numeric scalar; scaling for plot labels. Default is the value of this_plot_cex.
this_axis_cex	Numeric scalar; scaling for plot axis. Default is the value of this_plot_cex.
this_sub_cex	Numeric scalar; scaling for plot labels. Default is the value of 0.5.
this_mar	Numeric vector; set margins for the plot. Default is c(4, 4, 4, 0, 5).
sub_title_one_line	Integer scalar; position of first subtitle of plot. Default is 1.5.
sub_title_two_line	Integer scalar; position of main title of plot. Default is 0.5.
this_reset	Logical scalar; if TRUE, the values of par are reset. Default is TRUE.

Details

Version 1.4, 2026-05-21

Value

Generates a plot of the regARIMA residuals with diagnostic information in the sub-headers.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

NBER (2026). “Business Cycle Dating.” <https://www.nber.org/research/business-cycle-dating>.

R Coder (2025). “Colors in R — r-charts.com.” <https://r-charts.com/colors/>. [Accessed 17-02-2026].

Sax C, Eddelbuettel D (2018). “Seasonal Adjustment by X-13ARIMA-SEATS in R.” *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
air_seas <-
  seasonal::seas(AirPassengers, arima.model = '(0 1 1)(0 1 1)')
plot_resid(air_seas,
  main_title = 'ARIMA Residuals for Airline Passengers',
  use_ratio = TRUE, line_color='darkblue')
```

plot_series	<i>Plot individual series.</i>
-------------	--------------------------------

Description

Generate plot of user-specified series.

Usage

```

plot_series(
  this_series = NULL,
  main_title = NULL,
  main_title_line = 2.75,
  main_title_cex = 1.25,
  x_axis_label = NULL,
  y_axis_label = NULL,
  x_axis_main_ticks = NULL,
  y_axis_main_ticks = NULL,
  y_limit = NULL,
  start_plot = NULL,
  do_grid = FALSE,
  draw_recess = FALSE,
  recess_start = NULL,
  recess_col = "lightgrey",
  recess_sub = TRUE,
  this_trans = TRUE,
  use_ratio = FALSE,
  line_color = "grey",
  this_plot_type = "l",
  this_line_type = 1,
  this_line_width = 1,
  this_point_type = NULL,
  this_plot_cex = 0.8,
  this_lab_cex = NULL,
  this_axis_cex = NULL,
  this_mar = c(4, 4, 4, 0.5),
  this_reset = TRUE
)

```

Arguments

<code>this_series</code>	Numeric vector; time series object to be plotted. This is a required entry.
<code>main_title</code>	Character string; main title of plot. Default is no title.
<code>main_title_line</code>	Integer scalar; position of main title of plot. Default is 2.75.
<code>main_title_cex</code>	Numeric scalar; scaling for main title of plot. Default is 1.25.
<code>x_axis_label</code>	Character string; x-axis label for plot, if specified.
<code>y_axis_label</code>	Character string; y-axis label for plot, if specified.
<code>x_axis_main_ticks</code>	Numeric vector; X-axis tick marks. Default is determined by the <code>axis</code> function.
<code>y_axis_main_ticks</code>	Numeric vector; Y-axis tick marks. Default is determined by the <code>axis</code> function.
<code>y_limit</code>	Numeric vector of length 2; Range of values you wish the plot to be plotted over. Default is range of <code>this_series</code> .
<code>start_plot</code>	Integer vector of length 2; Starting date for plot. Default is starting date for the time series.
<code>do_grid</code>	Logical scalar; indicates if certain plots will have grid lines. Default is no grid lines.

draw_recess	Logical scalar; indicates if certain plots will have shaded areas for NBER recession dates (NBER 2026). Default is no recession shading.
recess_start	numeric matrix; Rows of dates for additional recession starting and ending dates. Default is not to add recession dates.
recess_col	Character string; color used for shading of recession region. Default is "lightgrey". For a listing of R color codes, see R Coder (2025).
recess_sub	Logical scalar; indicates if x-axis label for recession is produced for this plot. Default is x-axis label is produced
this_trans	Logical scalar; indicates if the adjustment was done with a log transform. Default is TRUE.
use_ratio	Logical scalar; indicates if plots of seasonal factors, irregular, and residuals are done as ratio plots. Default has these plots as time series line plots.
line_color	Character string; color used for series. Default is 'grey'. For a listing of R color codes, see R Coder (2025).
this_plot_type	Character string; type of plot produced. Typical values are "l" (lines only), "p" (points only), "b" (both points and lines), "o" (overplotted points and lines), "n" (no plotting). Default is "l".
this_line_type	Integer scalar; line type used for series. Default is 1.
this_line_width	Integer scalar; line width used for series. Default is 1.
this_point_type	Integer scalar; point type used for series. Default is no points plotted.
this_plot_cex	Numeric scalar; scaling for the plot itself. Default is 0.8.
this_lab_cex	Numeric scalar; scaling for plot labels. Default is the value of this_plot_cex.
this_axis_cex	Numeric scalar; scaling for plot axis. Default is the value of this_plot_cex.
this_mar	Numeric vector; set margins for the plot. Default is c(4, 4, 4, 0, 5).
this_reset	Logical scalar; if TRUE, the values of par are reset. Default is TRUE.

Details

Version 2.0, 2026-04-08

Value

Generate plot of user-specified series. Can be first in a series of plots, with other lines or points added after calling this routine. If series not specified, print out error message and return NULL.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

NBER (2026). "Business Cycle Dating." <https://www.nber.org/research/business-cycle-dating>.

R Coder (2025). "Colors in R — r-charts.com." <https://r-charts.com/colors/>. [Accessed 17-02-2026].

Examples

```
air_seas <-
  seasonal::seas(AirPassengers,
    arima.model = "(0 1 1)(0 1 1)",
    outlier.types = "all",
    x11 = "",
    forecast.maxlead = 36)
air_sa <- seasonal::final(air_seas)
plot_series(air_sa,
  y_axis_label = "AirPass",
  do_grid = TRUE,
  draw_recess = TRUE,
  line_color = "black",
  start_plot = c(1958,1),
  this_point_type = 1,
  main_title = "X-11 Seasonal Adjustment for Airline Passengers",
  main_title_line = 1.5,
  main_title_cex = 1.0,
  this_reset = TRUE)
```

plot_sf

Seasonal factor (and the SI-ratios) plot grouped by month/quarter

Description

Generates a special plot of the seasonal factors (and the SI-ratios) grouped by month/quarter

Usage

```
plot_sf(
  seas_obj = NULL,
  this_table = NULL,
  add_si = FALSE,
  main_title = "Seasonal Sub-Plots",
  sub_title = NULL,
  y_label = NULL,
  y_limit = NULL,
  x_label = " ",
  line_color = NULL,
  this_plot_cex = 0.8,
  this_lab_cex = NULL,
  this_main_cex = NULL,
  this_axis_cex = NULL,
  this_mar = c(2.1, 1.5, 3.1, 0.5),
  num_facet_rows = NULL,
  do_grid = FALSE,
  this_facet_bg = "grey90",
  this_facet_cex = 0.6,
  this_tinytheme = "minimal",
  this_reset = TRUE
)
```

Arguments

seas_obj	A seas object for a single series generated from the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.
this_table	Character string; X-13ARIMA-SEATS table name or abbreviation (such as d10 or s10). Valid table names for this argument can be found in the documentation for the series function of the seasonal package (Sax 2018). If not a valid table name, the function will print an error message and return a NULL. This entry is required.
add_si	Logical scalar; indicates if seasonal factor plots will include the SI ratios and unmodified SI ratios for X-11 seasonal adjustments. Default is not including SI ratios (add_si = FALSE).
main_title	Character string; main title of plot. Default is 'Seasonal Sub-Plots'.
sub_title	Character string; a subtitle for the plot. Default is NULL, which generates a subtitle which denotes the series that are plotted.
y_label	Character string; y-axis label for plot, if specified.
y_limit	Numeric vector of length 2; Range of values to be plotted over. Default is range of the seasonal factors.
x_label	Character string; label for x-axis of plot. Default is a blank x-axis.
line_color	Character array of length 2 or 4; color used for seasonal factors, SI-ratios, Modified SI-ratios, and seasonal means. Default is generated from the RColorBrewer palette "Dark2", with "darkgrey" added for the seasonal mean line. For a listing of R color codes, see R Coder (2025).
this_plot_cex	Numeric scalar; scaling for the plot itself. Default is 0.8.
this_lab_cex	Numeric scalar; scaling for plot labels. Default is the value of this_plot_cex.
this_main_cex	Numeric scalar; scaling for main plot title. Default is the value of this_plot_cex + 0.1.
this_axis_cex	Numeric scalar; scaling for plot axis. Default is the value of this_plot_cex.
this_mar	Numeric vector; set margins for the plot. Default is c(2.1, 1.5, 3.1, 0.5).
num_facet_rows	Integer scalar; number of rows in the facet plot. Default is generated from the series frequency (2 for monthly series, 1 otherwise).
do_grid	Logical scalar; indicates if certain plots will have grid lines. Default is no grid lines.
this_facet_bg	Character scalar; Color used for the facet labels. Default is "grey90".
this_facet_cex	Integer scalar; size used for the facet labels. Default is 0.6.
this_tinytheme	Character scalar; setting for the tinythere function from the tinypplot package (McDermott et al. 2025). Default is "minimal".
this_reset	Logical scalar; if TRUE, the values of par are reset. Default is TRUE.

Details

Version 1.3.1, 2026-05-14

Value

Generates a special plot of the seasonal factors (and the SI-ratios) grouped by month/quarter.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

McDermott G, Arel-Bundock V, Zeileis A (2025). *tinypilot: Lightweight Extension of the Base R Graphics System*. doi:10.32614/CRAN.package.tinypilot, R package version 0.6.0.

R Coder (2025). “Colors in R — r-charts.com.” <https://r-charts.com/colors/>. [Accessed 17-02-2026].

Sax C (2018). *seasonal: R Interface to X-13ARIMA-SEATS*. R package version 1.7. <https://CRAN.R-project.org/package=seasonal>.

Sax C, Eddelbuettel D (2018). “Seasonal Adjustment by X-13ARIMA-SEATS in R.” *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
air_seas <- seasonal::seas(AirPassengers, arima.model = '(0 1 1)(0 1 1)',
                           x11='', x11.save = c("d8", "d9"))
plot_sf(air_seas, add_si = TRUE,
        main_title = 'Air Passengers Seasonal Sub-Plots',
        sub_title = "SF = Cyan, SI = Orange, ModSI = Purple, Mean = Gray",
        do_grid = TRUE)
```

plot_table

Plot table from X-13ARIMA-SEATS seasonal adjustment run.

Description

Generate plot of user-specified table from the output of a seas object generated by the seasonal package.

Usage

```
plot_table(
  seas_obj = NULL,
  this_table = NULL,
  main_title = NULL,
  main_title_line = 2,
  main_title_cex = 1.25,
  x_axis_label = NULL,
  y_axis_label = NULL,
  x_axis_main_ticks = NULL,
  y_axis_main_ticks = NULL,
  y_limit = NULL,
  start_plot = NULL,
  do_grid = FALSE,
  draw_recess = FALSE,
  recess_start = NULL,
```

```

    recess_col = NULL,
    recess_sub = TRUE,
    add_otl = FALSE,
    use_ratio = FALSE,
    add_sub_title = FALSE,
    sub_title_line = 1,
    sub_title_cex = NULL,
    this_line_type = NULL,
    this_line_width = 1,
    this_point_type = NULL,
    line_color = "grey",
    otl_col = NULL,
    this_plot_cex = 0.8,
    this_lab_cex = NULL,
    this_axis_cex = NULL,
    this_mar = c(4, 4, 4, 0.5),
    this_reset = FALSE
  )

```

Arguments

seas_obj	A seas object for a single series generated from the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.
this_table	Character string; X-13ARIMA-SEATS table name or abbreviation. If not a valid table name, the function will print an error message and return a NULL. This entry is required.
main_title	Character string; main title of plot. Default is 'Cumulative periodogram'.
main_title_line	Integer scalar; position of main title of plot. Default is 2.75.
main_title_cex	Numeric scalar; scaling for main title of plot. Default is this_plot_cex + 0.1.
x_axis_label	Character string; x-axis label for plot, if specified.
y_axis_label	Character string; y-axis label for plot, if specified.
x_axis_main_ticks	Numeric vector; X-axis tick marks. Default is determined by the axis function.
y_axis_main_ticks	Numeric vector; Y-axis tick marks. Default is determined by the axis function.
y_limit	Numeric vector of length 2; Range of values you wish the plot to be plotted over. Default is range of the series specified.
start_plot	Integer vector of length 2; Starting date for plot. Default is starting date for the time series.
do_grid	Logical scalar; indicates if certain plots will have grid lines. Default is no grid lines.
draw_recess	Logical scalar; indicates if certain plots will have shaded areas for NBER recession dates (NBER 2026). Default is no recession shading.
recess_start	numeric matrix; Rows of dates for additional recession starting and ending dates. Default is not to add recession dates.
recess_col	Character string; color used for shading of recession region. Default is 'lightgrey'.
recess_sub	Logical scalar; indicates if x-axis label for recession is produced for this plot. Default is x-axis label is produced

add_otl	Logical scalar; indicates if lines for identified outliers are included in series plots. Default is not including lines for identified outliers.
use_ratio	Logical scalar; indicates if plots of seasonal factors, irregular, and residuals are done as ratio plots. Default has these plots as time series line plots.
add_sub_title	Logical scalar; indicates if plots will include subtitles denoting what series are plotted. Default is not including subheaders.
sub_title_line	Integer scalar; position of main title of plot. Default is 1.
sub_title_cex	Numeric scalar; scaling for main title of plot. Default is <code>this_plot_cex - 0.1</code> .
this_line_type	Integer scalar; indicates line type of plot. Default is 1.
this_line_width	Integer scalar; line width used for series. Default is 1.
this_point_type	Integer scalar; point type used for series. Default is no points plotted.
line_color	Character scalar; color used for plot. Default is 'grey'. For a listing of R color codes, see R Coder (2025).
otl_col	Character array of length 6; color used for different outliers, with the order being 'ao', 'ls', 'tc', 'so', 'rp', 'tls'. Default is colors from the RColorBrewer palette "Paired" (Neuwirth 2022).
this_plot_cex	Numeric scalar; scaling for the plot itself. Default is 0.8.
this_lab_cex	Numeric scalar; scaling for plot labels. Default is the value of <code>this_plot_cex</code> .
this_axis_cex	Numeric scalar; scaling for plot axis. Default is the value of <code>this_plot_cex</code> .
this_mar	Numeric vector; set margins for the plot. Default is <code>c(4, 4, 4, 0, 5)</code> .
this_reset	Logical scalar; if TRUE, the values of <code>par</code> are reset. Default is FALSE.

Details

Version 1.6, 2026-07-08

Value

Generate plot of user-specified series. If series not specified, print out error message and return NULL.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

- NBER (2026). "Business Cycle Dating." <https://www.nber.org/research/business-cycle-dating>.
- Neuwirth E (2022). *RColorBrewer: ColorBrewer Palettes*. R package version 1.1-3, <https://CRAN.R-project.org/package=RColorBrewer>.
- R Coder (2025). "Colors in R — r-charts.com." <https://r-charts.com/colors/>. [Accessed 17-02-2026].
- Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
air_seas <-
  seasonal::seas(AirPassengers,
    arima.model = "(0 1 1)(0 1 1)",
    x11="", x11.save = c("e18"))
blstinplot::plot_table(air_seas, "d11",
  y_axis_label = "AirPassengers",
  main_title = "X-11 Seasonal Adjustment of Airline Passengers",
  main_title_cex = 1.0,
  add_sub_title = TRUE,
  do_grid = TRUE,
  draw_recess = TRUE,
  use_ratio = FALSE,
  add_otl = TRUE,
  line_color = "black")
blstinplot::plot_table(air_seas, "e18",
  y_axis_label = "AirPassengers",
  main_title = "Adjustment Factors of Airline Passengers",
  main_title_cex = 1.0,
  add_sub_title = TRUE,
  do_grid = TRUE,
  draw_recess = TRUE,
  use_ratio = TRUE,
  add_otl = TRUE,
  line_color = "darkblue")
```

plot_year_over_year	<i>Year over year plot of individual series.</i>
---------------------	--

Description

Generate plot of user-specified series with each year as a separate line.

Usage

```
plot_year_over_year(
  this_series = NULL,
  main_title = NULL,
  main_title_line = 1.75,
  main_title_cex = NULL,
  x_label_line = 2.5,
  y_limit = NULL,
  do_grid = FALSE,
  line_color = NULL,
  this_palette = "Paired",
  detrend_series = FALSE,
  detrend_lowess = FALSE,
  start_plot = NULL,
  two_digit_year = TRUE,
  this_legend = "direct",
  this_legend_cex = 0.75,
  this_plot_cex = 0.8,
```

```

    this_lab_cex = NULL,
    this_axis_cex = NULL,
    this_mar = c(3, 3, 3, 0.5),
    this_reset = TRUE
  )

```

Arguments

<code>this_series</code>	Numeric vector; time series object to be plotted.
<code>main_title</code>	Character string; main title of plot. Default is no title.
<code>main_title_line</code>	Integer scalar; position of main title of plot. Default is 1.75.
<code>main_title_cex</code>	Numeric scalar; scaling for main title of plot. Default is <code>this_plot_cex + 0.1</code> .
<code>x_label_line</code>	Integer scalar; position of X-axis label of plot. Default is 2.5.
<code>y_limit</code>	Numeric vector of length 2; Range of values you wish the plot to be plotted over. Default is range of the series specified.
<code>do_grid</code>	Logical scalar; indicates if plots will have grid lines. Default is no grid lines.
<code>line_color</code>	Character scalar; color used for plot. User should specify one color for each column of the matrix specified. Default is the RColorBrewer palette "Paired". For a listing of R color codes, see R Coder (2025).
<code>this_palette</code>	Character string; RColorBrewer palette used for this plot. Default is the RColorBrewer palette "Paired" (Neuwirth 2022).
<code>detrend_series</code>	Logical scalar; indicates if the series plotted is to be detrended. Default is the original series is plotted.
<code>detrend_lowess</code>	Logical scalar; indicates lowess is used to generate the trend used to detrend the series. Default setting is loess is not used to detrend the series; detrending doesn't happen unless <code>detrend_series = TRUE</code> .
<code>start_plot</code>	Integer vector of length 2; Starting date for plot. Default is starting date for the time series.
<code>two_digit_year</code>	Logical scalar; indicates if two digit representation of the year is used in the plot. Default is <code>two_digit_year = TRUE</code> .
<code>this_legend</code>	Character string; sets the position of the plot legend. Default is "direct", which labels each line with the year it represents.
<code>this_legend_cex</code>	Numeric scalar; scaling for the legend. Default is 0.75. Not used if <code>this_legend = "direct"</code> .
<code>this_plot_cex</code>	Numeric scalar; scaling for the plot itself. Default is 0.8.
<code>this_lab_cex</code>	Numeric scalar; scaling for plot labels. Default is the value of <code>this_plot_cex</code> .
<code>this_axis_cex</code>	Numeric scalar; scaling for plot axis. Default is the value of <code>this_plot_cex</code> .
<code>this_mar</code>	Numeric vector; set margins for the plot. Default is <code>c(3, 3, 3, 0.5)</code> .
<code>this_reset</code>	Logical scalar; if TRUE, the values of <code>par</code> are reset. Default is TRUE.

Details

Version 2.1, 2026-07-10

Value

Generate year over year plot of user-specified series. If series not specified, print out error message and return NULL.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Neuwirth E (2022). *RColorBrewer: ColorBrewer Palettes*. R package version 1.1-3, <https://CRAN.R-project.org/package=RColorBrewer>.

R Coder (2025). “Colors in R — r-charts.com.” <https://r-charts.com/colors/>. [Accessed 17-02-2026].

Examples

```
plot_year_over_year(AirPassengers,
  main_title = "Airline Passenger Series (1949 - 1960)")
```

proc_outlier	<i>Extract dates from outlier text</i>
--------------	--

Description

Process name of outlier regressor to extract the dates associated with the outlier.

Usage

```
proc_outlier(this_outlier = NULL, this_freq = 12, add_type = TRUE)
```

Arguments

this_outlier	Character string; outlier regressor. This is a required entry.
this_freq	integer scalar; time series frequency. Default is 12.
add_type	logical scalar; determines if type of outlier is added to the output. Default is TRUE.

Details

Version 2.2, 2026-02-26

Value

List of either year and month/quarter of outlier, or year and month/quarter of start and end of outlier

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

Examples

```
air_seas <-
  seasonal::seas(AirPassengers,
    transform.function = "log",
    arima.model = "(0 1 1)(0 1 1)",
    regression.aictest = 'td',
    forecast.maxlead=36,
    check.print = c( "pacf", "pacfplot" ),
    x11="",
    slidingspans = "",)
this_auto_outlier <- get_auto_outlier_string(air_seas)
this_outlier      <- proc_outlier(this_auto_outlier)
```

reset_par	<i>Reset par()</i>
-----------	--------------------

Description

Reset graphics parameters for plots; taken from Stack Overflow (2012).

Usage

```
reset_par(old_par = NULL)
```

Arguments

old_par List object; old graphical parameters saved from par. This is a required entry.

Details

Version 1.1, 2026-04-07

Value

returns Resets R graphics parameters.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Stack Overflow (2012). "Reset the graphical parameters back to default values without use of dev.off()." [Accessed 01-04-2026], <https://stackoverflow.com/questions/9292563/reset-the-graphical-par>

Examples

```
old.par <- par(no.readonly = TRUE)
par(mar=c(5.1, 3.1, 4.1, 1.1), mfrow=c(2,2))
xt_names <- names(xt_data_list)
for (i in 1:4) {
  plot(xt_data_list[[i]], main = xt_names[i], type="l")
}
reset_par(old.par)
```

set_day_and_month	<i>generate date object from days and year</i>
-------------------	--

Description

Generate a date object from the number of days from the beginning of the year and the target year. Uses the lubridate package Grolemund and Wickham (2011)).

Usage

```
set_day_and_month(day_number, target_year)
```

Arguments

day_number	Integer scalar, number of days from the start of the year. This is a required entry.
target_year	Integer scalar, year of the date object. This is a required entry.

Details

Version 1.1.1, 2026-06-18

Value

Date object determined by the input arguments.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

Examples

```
this_start <- set_day_and_month(1, 2001)
this_end   <- set_day_and_month(350, 2025)
```

shoes2008	<i>Retail sales of shoes, 2008</i>
-----------	------------------------------------

Description

A time series object containing retail sales of shoes from the US Census Bureau.

Usage

```
data(shoes2008)
```

Format

Retail sales of shoes ending in April of 2008.

visual_sig_peaks	<i>Flag visual significant peaks in spectra.</i>
------------------	--

Description

Determine positions of visual significant peaks in spectra using the criteria of Soukup and Findley (1999).

Usage

```
visual_sig_peaks(seas_obj = NULL, spec_type = "sa", spec_freq_code = "seas")
```

Arguments

seas_obj	A seas object for a single series generated from the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.
spec_type	Character string; type of spectrum. Possible values are 'ori', 'irr', 'rsd', 'sa', 'comp', 'indsa', 'indirr', 'extrsd'. Default is 'sa'.
spec_freq_code	Character string; type of frequency being tested. Possible values are 'seas' or 'td'. Default is 'seas'.

Details

Version 1.1, 2026-06-03

Value

If visually significant peaks found, a numeric vector of the position of the peak frequencies. If no peaks found, 0 is returned.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Sax C, Eddelbuettel D (2018). “Seasonal Adjustment by X-13ARIMA-SEATS in R.” *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Soukup RJ, Findley DF (1999). “On the Spectrum Diagnostics Used by X-12-ARIMA to Indicate the Presence of Trading Day Effects after Modeling or Adjustment.” *Proceeding of the American Statistical Association, Business and Economic Statistics Section*, 144–149. <https://www.census.gov/content/dam/Census/library/working-papers/1999/adrm/rr9903s.pdf>.

Examples

```
air_seas <-
  seasonal::seas(AirPassengers,
    arima.model = '(0 1 1)(0 1 1)',
    x11='')
vp_ori_seas <-
  visual_sig_peaks(air_seas, spec_type = 'ori')
```

```
vp_ori_td <-
  visual_sig_peaks(air_seas,
    spec_type = 'sa',
    spec_freq_code = 'td')
```

xt_data_list

US Building Permits, One Family Units

Description

A list object of US One family Building Permits from the US Census Bureau for four regions expressed as time series objects that end in October, 2006.

Usage

```
data(xt_data_list)
```

Format

A list object with 4 time series elements:

mw1u Midwest one family building permits

ne1u Northeast one family building permits

so1u South one family building permits

we1u West one family building permits

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